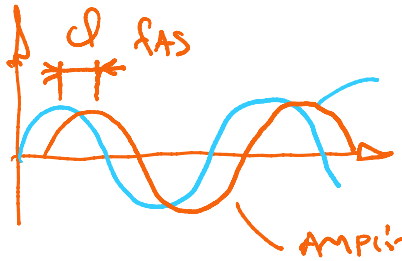
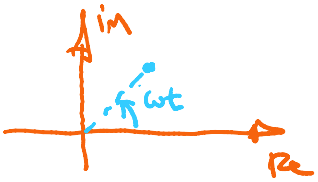
den 28 september 2021 10:05

LEKTION 12.



φ TALAR OM HUR STOR SKILLNAD DET ÄR MELLAN TVÅ KURVOR (MED SAMMA ω)

$$\begin{cases} V(t) = V_0 \sin(\omega t) \\ i(t) = I_0 \sin(\omega t + \varphi) \end{cases}$$ ~ FASSKILLNAD MELLAN STRÖM & SPÄNNING



$V(t) = V_0 e^{j\omega t + 0^\circ}$ → FAS SOM SÄTTER IGÅNG

$V(t) = V_0 \angle 0^\circ$ ← FASVINKEL

AMPLITUD

$I(t) = I_0 \angle \varphi$ ←

KALLAS VINKELHAKE NOTATION ELLER STEINMETZ NOTATION

RÄKNEREGLER:

$$\begin{cases} Z_1 = A \angle \varphi_1 \\ Z_2 = B \angle \varphi_2 \end{cases}$$

- $\frac{Z_1}{Z_2} = \frac{A \angle \varphi_1}{B \angle \varphi_2} = \frac{A}{B} \angle \varphi_1 - \varphi_2$
- $Z_1 \cdot Z_2 = A \angle \varphi_1 \cdot B \angle \varphi_2 = AB \angle \varphi_1 + \varphi_2$
- $Z_1 + Z_2 = A \angle \varphi_1 + B \angle \varphi_2 =$

$$= A(\cos\varphi_1 + j\sin\varphi_1) + B(\cos\varphi_2 + j\sin\varphi_2) =$$

$$= \underbrace{A\cos\varphi_1 + B\cos\varphi_2}_{\text{NY REALDEL} = C} + j\underbrace{(A\sin\varphi_1 + B\sin\varphi_2)}_{\text{NY IM. DEL} = D}$$

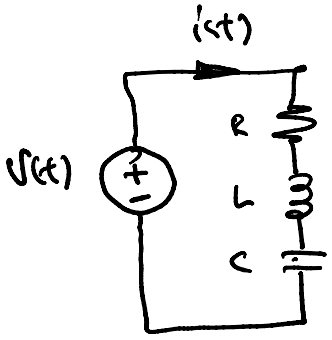
φ_2

$= C$ $= D$

$$= C + jD = \underbrace{\sqrt{C^2+D^2}}_{E} \angle \arctan\left(\frac{D}{C}\right) = \varphi_3$$

$$= E \angle \varphi_3$$

I går gjorde vi en beräkning av en ersättningsimpedans



$R = 25\ \Omega$
$L = 20\ mH$
$C = 50\ nF$
$v(t) = 50 \cos(\omega t + 30^\circ)\ V$
$f = 60\ Hz.$

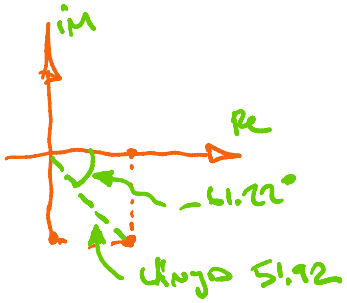
Beräkna strömmen $i(t)$ i kretsen
$Z_{ERS} = 25 - 45.51j\ \Omega$

Lösning: Vi vet att $I = \frac{V}{Z_{ERS}}$

där $V = 50 \angle 30^\circ$ volt.

Då blir $I = \frac{V}{Z_{ERS}} = \frac{50 \angle 30^\circ}{25 - 45.51j} =$

$$= \frac{50 \angle 30^\circ}{51.92 \angle -61.22^\circ} = 0.96 \angle 91.22^\circ \ A$$



$\sqrt{25^2 + 45.51^2} = 51.92$

$\arctan\left(\frac{-45.51}{25}\right) = -61.22^\circ$

Men vi måste ta oss från w-planet till tidsplanet

gäller inte ejwt!

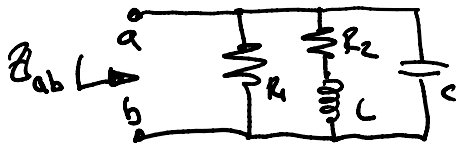
$$i(t) = \mathrm{Re}\{I\} = \mathrm{Re}\{0.96 \angle 91.22^\circ\}$$

$= 0.96 \sin(\omega t + 91.22°)$

SVAR: $i(t) = 0.96 \sin(\omega t + 91.22°)$

$= 0.96 \cos(\omega t + 1.22°)$

(4.47) EX. BESTÄM EKV. IMPEDANS FÖR KOPPLINGEN NEDAN.



$R_1 = 800\Omega$
$R_2 = 500\Omega$
$L = 200\ mH$
$\omega = 4000\ RAD/SEK$

LÖSNING:

$$\mathcal{Z}_{ab}^{-1} = \frac{1}{R_1} + \frac{1}{R_2 + \mathcal{Z}_L} + \frac{1}{\mathcal{Z}_C}$$

$$\begin{cases} \mathcal{Z}_L = j\omega L \\ \mathcal{Z}_C = \frac{1}{j\omega C} \end{cases}$$

$$\mathcal{Z}_{ab}^{-1} = \frac{1}{R_1} + \frac{1}{R_2 + j\omega L} + j\omega C$$

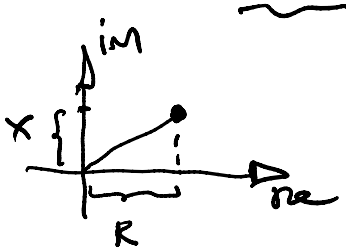
STOPPA IN VÄRDEN ... EFTER LITE ALGEBRA

$$\mathcal{Z}_{ab}^{-1} = \frac{1120.8 + j\,912}{4\cdot10^5 + j\,64\cdot10^5} = \frac{1445\angle 39.1°}{(7.56\angle 54.99°)\cdot 10^5}$$

$$\Rightarrow \mathcal{Z}_{ab} = 522.28\angle 18.86°\ \Omega$$

$$\mathcal{Z}_{EKV} = 522.28\angle 18.86°\ \Omega$$

SVAR: $\mathcal{Z}_{EKV} \approx 522\angle 19°$



$\mathcal{Z} = R + jX$

RESISTIV DEL ≥ 0 ALLTID

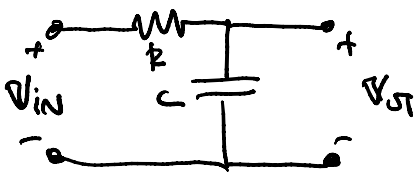
REAKTIV DEL (REAKTANS)

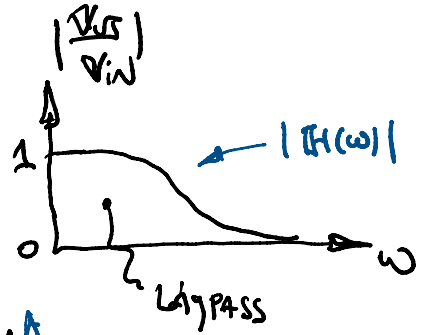
$\begin{cases} -X & \text{KAN VARA} \\ +X & \text{ANTINGEN} \end{cases}$

ELLER

ALLTID { + X ANTINGEN ELLER

FILTER:

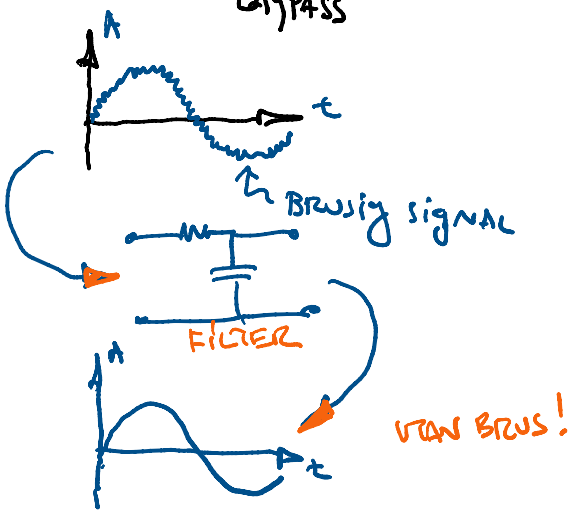


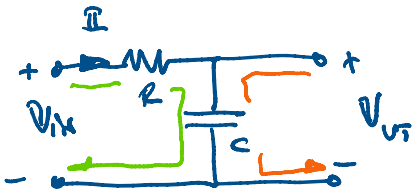


$\frac{V_{UT}}{V_{IN}} = f(\omega)$

ÖVERFÖRINGSFUNKTION

$\frac{V_{UT}}{V_{IN}} = H(\omega)$





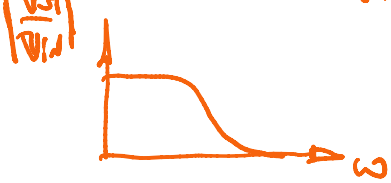
KRETSLAGEN

$\begin{cases} -V_{IN} + R\,I + Z_C\,I = 0 & (1) \\ -V_{UT} + Z_C\,I = 0 & (2) \end{cases}$

ELIMINERA I MELLAN (1) & (2)

$$\frac{V_{UT}}{V_{IN}} = \frac{Z_C}{R + Z_C} = \frac{\frac{1}{j\omega C}}{R + \frac{1}{j\omega C}} = \frac{1}{1 + j\omega CR}$$

$$\left|\frac{V_{UT}}{V_{IN}}\right| = \left|\frac{1}{1 + j\omega CR}\right| = \frac{1}{\sqrt{1 + (\omega CR)^2}}$$



EN ANNAN PARAMETER MAN VILL VETA ÄR HUR UTSPÄNNING FÖRHÅLLER SIG TILL INSPÄNNINGEN I FAS:

$$\frac{V_{UT}}{V_{IN}} = \frac{1}{1 + j\omega RC} = Z$$

$$\frac{V_{UT}}{V_{IN}} = \frac{1}{1+j\omega RC} = \mathcal{H}$$

$$\phi = \arctan\left(\frac{\mathrm{im}(\mathcal{H})}{\mathrm{re}(\mathcal{H})}\right)$$

ϕ = FASSKILLNAD MELLAN V_{UT} OCH V_{IN}

$$\frac{V_{UT}}{V_{IN}} = \frac{1}{1+j\omega RC} = \frac{1-j\omega RC}{(1+j\omega RC)(1-j\omega RC)} =$$

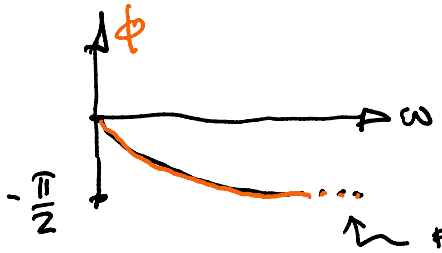
$$= \frac{1-j\omega RC}{1+(\omega RC)^2}$$

... = REALDEL

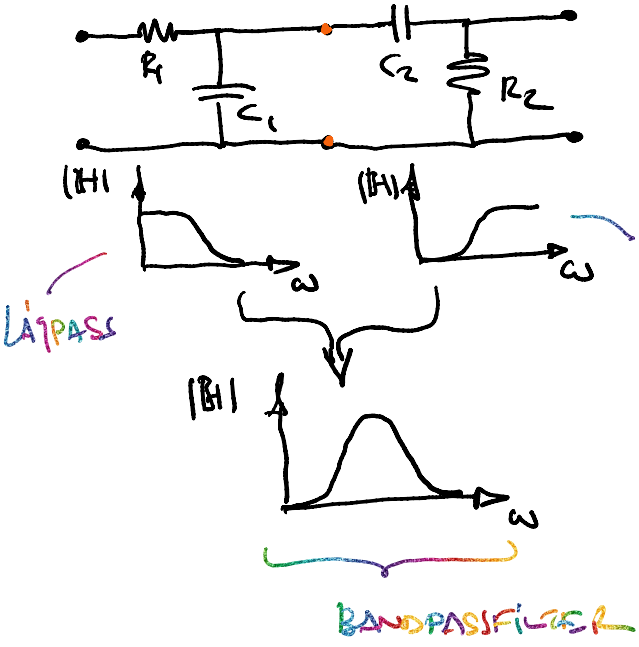
... = IMAGINÄRDEL

$$\phi = \arctan\left(\frac{-\omega RC}{1}\right) = -\arctan(\omega RC)$$

↑ FASFUNKTION



↑ FÖR HÖGA FREKV. FÅR FASFÖRSKJUTN. -90° MOT INSPÄNNINGEN

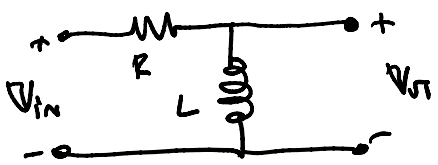


TVÅ FILTER EFTER VARANDRA (KASKADKOPPLADE)

FILTER 19EN

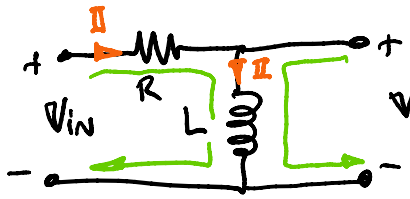


Beskriv $H(\omega)$ för filtret.



for filter.

Løsning:



vi får med KVL

$$\begin{cases} -V_{in} + RI + j\omega L I = 0 \\ -V_{ut} + j\omega L I = 0 \quad (2) \end{cases}$$

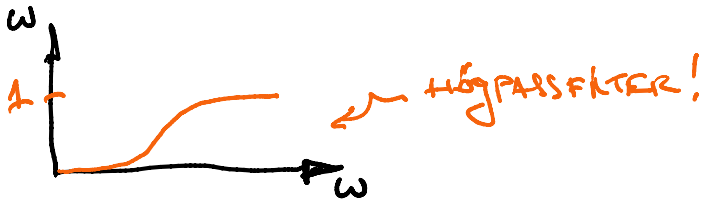
(2) giver $I = \dfrac{V_{ut}}{j\omega L}$

(1) giver

$$-V_{in} + (R + j\omega L)\frac{V_{ut}}{j\omega L} = 0$$

$$\frac{V_{ut}}{V_{in}} = \frac{j\omega L}{R + j\omega L} = H(\omega)$$

$$|H(\omega)| = \frac{\omega L}{\sqrt{R^2 + (\omega L)^2}}$$



SLUT!